

Activation-Trajectory Divergence for Machine Unlearning

ESIT-D2I 2026 Task 2 — CSI Indoor Localization

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3rd place — Public LB 0.92787 (ensemble)

Compliant single-model: offline LR $\sim$ acc **0.8475** / **0.8442**

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Task & Mandatory Metric

Task & mandatory evaluation

Problem

A pretrained CNN f_θ was trained on data containing a contaminated subset $\mathcal{F}$ (is_forget=1). **Goal:** remove $\mathcal{F}$'s influence *without retraining from scratch*, preserving accuracy on the clean retain set $\mathcal{R}$ and the unseen test set.

Mandatory evaluation scheme (2026-06-17 clarification)

$$\underbrace{\text{one fine-tuned model}}_{\text{must start from baseline}} \xrightarrow{\text{train}} \underbrace{e_i = \|f_\theta(x_i) - \mathbf{y}_i\|_2}_{\text{vs. original (corrupted) labels}} \xrightarrow{\text{LR}} \hat{z} \in \{0, 1\}$$

Single model only. No ensembles, no GMM thresholds, no OOF cross-fitting.

Key insight: $\mathbb{E}_{\mathcal{F}}[e_i] \gg \mathbb{E}_{\mathcal{R}}[e_i] \iff \text{good unlearning} \iff \text{high LR accuracy}.$

Data & Preprocessing

Dataset: DICHASUS Arena2036 CSI

Train set:

- $N = 10,918$ samples
- Retain ($z = 0$): 6,823 ●
- Forget ($z = 1$): 4,095 ●
- Predict 2-D position $(x, y) \in \mathbb{R}^2$
($z \approx 0.94$ m, constant)

Raw CSI shape:

$(N, 4, 2, 4, 64)$
samples $\times$ AP $\times$ pol $\times$ ant $\times$ sub

dtype: complex64

Test set: 2,728 samples; test positions available
(task2_test_positions.npy).

Preprocessing pipeline (`format_csi_for_cnn`)

Function: `src/dataset.py :: format_csi_for_cnn`

1. **Flatten antenna dims:** $(N, 4, 2, 4, 64) \longrightarrow (N, 32, 64)$
2. **Stack real and imaginary parts as two channels:**
 $(N, 32, 64) \longrightarrow (N, 2, 32, 64)$, dtype `float32`
3. **Global standardisation** (one scalar μ, σ over the *entire* array):

$$\hat{X} = \frac{X - \mu}{\sigma + 10^{-8}}, \quad \mu = \text{mean}(X), \quad \sigma = \text{std}(X)$$

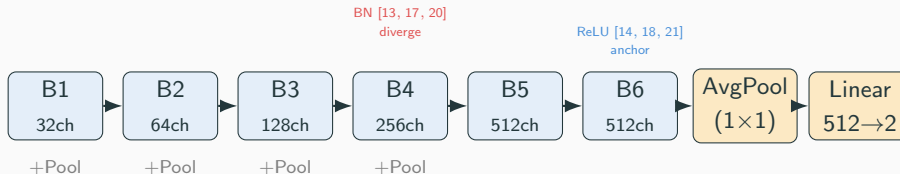
Matches the competition notebook exactly.

Positions: only (x, y) used; z dropped (`pos[:, :2]`).

Model Architecture

DichasusPositionPredictor

Input: $(B, 2, 32, 64)$ **Output:** $(B, 2)$ — predicted (x, y)



Each block: $\text{Conv2d}(k = 3, \text{pad} = 1) \rightarrow \text{BN} \rightarrow \text{ReLU}[\rightarrow \text{MaxPool}_{2 \times 2}]$.

Blocks 5–6 have no MaxPool. Hook index i addresses `model.features[i]` in the `nn.Sequential`.

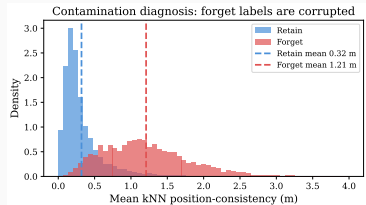
Contamination Diagnosis

What is the contamination?

Hypothesis: forget labels are *corrupted positions* — the CSI is genuine but the stored (x, y) is wrong.

kNN-consistency score: for sample i , find its 10 nearest retain-set neighbours in 64-dim PCA of $|\text{CSI}|$:

$$c_i = \frac{1}{K} \sum_{j \in \text{kNN}_{\mathcal{R}}(i)} \|\mathbf{p}_j - \mathbf{y}_i\|_2$$



	Retain	Forget
Mean c_i	0.32 m	1.21 m
AUC($z c$)	0.936	

$\Rightarrow$ Forget labels are **4 $\times$ less consistent** with their CSI.

The unlearning objective

The detector reads $e_i = \|f_\theta(x_i) - \mathbf{y}_i\|_2$ (vs. original labels). Good unlearning:

$$\mathbb{E}_{i \in \mathcal{F}}[e_i] \gg \mathbb{E}_{i \in \mathcal{R}}[e_i], \quad \mathbb{E}_{i \in \mathcal{R}}[e_i] \approx \text{const.}$$

Key observation: if the model predicts the *true* position, then for a forget sample with corrupted label $\mathbf{y}_i$:

$$e_i = \|f_\theta(x_i) - \mathbf{y}_i\|_2 \approx \underbrace{\|\mathbf{y}_i^{\text{true}} - \mathbf{y}_i\|_2}_{\text{corruption magnitude}}.$$

Error = corruption magnitude — a signal that transfers to test because the corruption is there too.

Probing the Baseline

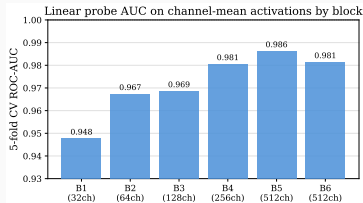
Does the CNN already “know” forget samples?

Procedure:

1. Hook post-ReLU outputs at blocks 1–6 (indices {2, 6, 10, 14, 18, 21}).
2. Compute channel-mean activations:

$$\bar{\mathbf{a}}_i^\ell = \frac{1}{HW} \sum_{h,w} a_{i,c,h,w}^\ell \in \mathbb{R}^{C_\ell}$$

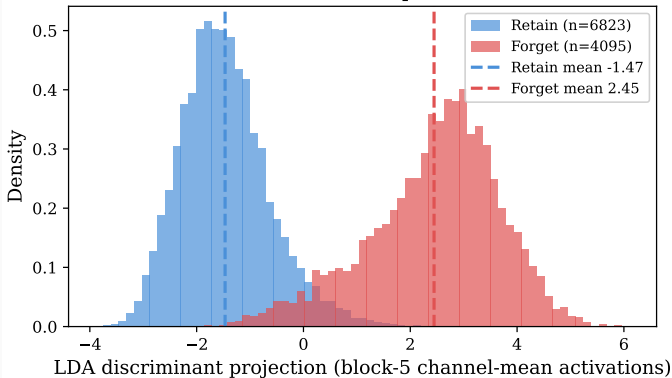
3. Fit LogReg + StandardScaler, 5-fold CV; report ROC-AUC.



Result: AUC rises **0.948** (B1) → **0.986** (B5). The CNN linearly separates forget from retain internally.

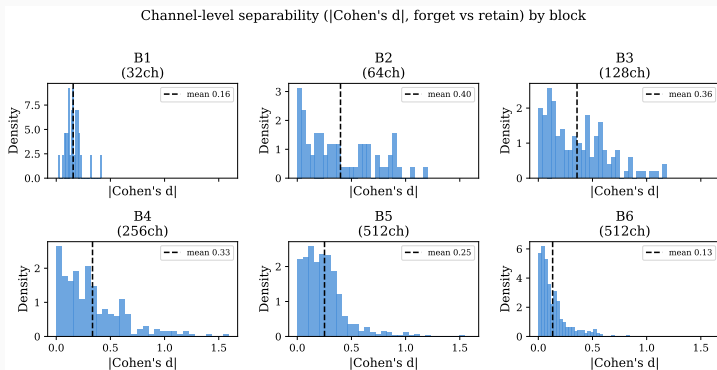
Activation-space separation (block 5)

Block-5 activations (baseline): forget / retain are linearly separable
LDA discriminant — 5-fold probe AUC = 0.986



LDA discriminant projection of 512-dim channel-mean activations at block 5 (baseline model). Forget (■) and retain (■) peaks are clearly separated along the single optimal discriminant direction $\Rightarrow$ unlearning should act in *activation space*.

Channel-level separability: Cohen's d



Pooled-std Cohen's $d = (\mu_{\mathcal{F}} - \mu_{\mathcal{R}}) / \sigma_{\text{pool}}$ per channel at each block. Separability deepens with depth; blocks 4–6 carry the strongest signal.

Fisher importance: where does forget influence live?

Diagonal empirical Fisher:

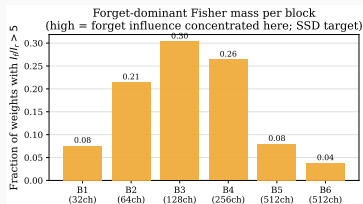
$$\mathcal{I}_{\theta} = \mathbb{E} \left[\left(\frac{\partial L_{\text{MSE}}}{\partial \theta} \right)^2 \right]$$

Computed separately on $\mathcal{F}$ (corrupted labels) and $\mathcal{R}$.

Fraction of weights with

$$\mathcal{I}^{\mathcal{F}} / \mathcal{I}^{\mathcal{R}} > 5:$$

Forget-dominant mass concentrates in **blocks 2–4**. Blocks 5–6 are less forget-specific despite having the most separable activations.



Negative result: SSD fails

Selective Synaptic Dampening: if

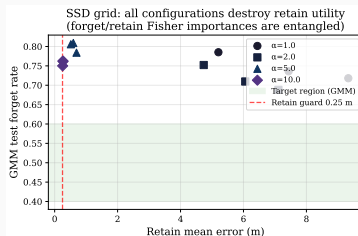
$\mathcal{I}_k^{\mathcal{F}} > \alpha \cdot \mathcal{I}_k^{\mathcal{R}}$, multiply weight k by $\min(\lambda \mathcal{I}_k^{\mathcal{R}} / \mathcal{I}_k^{\mathcal{F}}, 1)$. Grid:

$\alpha \in \{1, 2, 5, 10\}$, $\lambda \in \{0.1, 0.5, 1.0\}$.

BN recalibration on retain after dampening.

Green box = target region. **No point passes the retain guard.** Retain and forget errors collapse in lockstep.

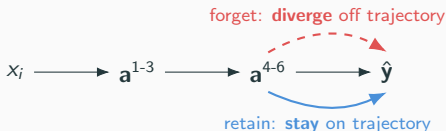
Conclusion: forget/retain Fisher importances are *entangled*. Must act in **activation space**.



Method: Activation-Trajectory Divergence

Method overview

Trajectory view: a forward pass is a trajectory through activation space. Frozen teacher f_{θ_0} defines the reference.



Design choices:

- *Where?* Blocks 4–6: highest separability, deepest representations.
- *Pre-ReLU BN* outputs for divergence: *signed* activations make the cosine push geometrically well-defined (no all-zero collapse shortcut).
- *Post-ReLU* for retain anchor: non-negative, shape-preserving.

Loss function

Student f_θ (trained), frozen teacher f_{θ_0} . Hooks at $\mathcal{L}_{\text{div}} = \{13, 17, 20\}$ (pre-ReLU BN) and $\mathcal{L}_{\text{anc}} = \{14, 18, 21\}$ (post-ReLU), blocks 4–6.

Total loss

$$\mathcal{L} = \underbrace{L_{\text{pos}}}_{\text{retain pos.}} + \beta \underbrace{L_{\text{anc}}}_{\text{retain anchor}} + \gamma_t \underbrace{L_{\text{div}}}_{\text{forget diverge}} + \nu \underbrace{L_{\text{norm}}}_{\text{norm-keeper}} + \delta \underbrace{L_{\text{pos,F}}}_{\text{forget pos.}}$$

Retain position MSE: $L_{\text{pos}} = \mathbb{E}_{i \in \mathcal{R}} \|f_\theta(x_i) - \mathbf{y}_i\|^2$

Retain anchor (relative squared deviation from teacher):

$$L_{\text{anc}} = \frac{1}{|\mathcal{L}_{\text{anc}}|} \sum_{\ell \in \mathcal{L}_{\text{anc}}} \mathbb{E}_{i \in \mathcal{R}} \left[\frac{\|\mathbf{a}_s^\ell - \mathbf{a}_t^\ell\|^2}{\|\mathbf{a}_t^\ell\|^2 + \varepsilon} \right]$$

Pins student retain activations to teacher's trajectory.

Loss function — forget terms

Forget diverge (hinged cosine at pre-ReLU BN layers):

$$L_{\text{div}} = \frac{1}{|\mathcal{L}_{\text{div}}|} \sum_{\ell \in \mathcal{L}_{\text{div}}} \mathbb{E}_{i \in \mathcal{F}} \left[\text{ReLU}(\cos(\mathbf{a}_s^\ell, \mathbf{a}_t^\ell) - m) \right], \quad m = 0$$

Penalises only *positive* cosine; saturates once orthogonal/anti-aligned.

Norm-keeper (blocks all-zero collapse):

$$L_{\text{norm}} = \frac{1}{|\mathcal{L}_{\text{div}}|} \sum_{\ell \in \mathcal{L}_{\text{div}}} \mathbb{E}_{i \in \mathcal{F}} \left[\left(\frac{\|\mathbf{a}_s^\ell\|_2 - \|\mathbf{a}_t^\ell\|_2}{\|\mathbf{a}_t^\ell\|_2 + \varepsilon} \right)^2 \right]$$

kNN-corrected forget position loss (--forget-target knn):

$$L_{\text{pos,F}} = \mathbb{E}_{i \in \mathcal{F}} \|f_\theta(x_i) - \hat{\mathbf{y}}_i^{\text{corr}}\|^2$$

$\gamma_t = \gamma \cdot \min(1, \text{epoch}/2)$ warms up over 2 epochs. Hyperparameters:
 $\beta = 3, \gamma = 0.5, \nu = 0.1, \delta = 1.$

kNN-corrected forget positions

Why needed: the cosine divergence pushes forget activations off-trajectory but does not specify *where* they should land. Without a position target the model has no incentive to produce a large error at test time.

Algorithm (src/dataset.py :: knn_corrected_positions):

1. Compute $|\text{CSI}|$ magnitudes, flatten to $(N, 2048)$.
2. Reduce to 64-dim PCA features.
3. For each forget sample i : find $k = 10$ nearest retain neighbours;

$$\hat{\mathbf{y}}_i^{\text{corr}} = \frac{1}{k} \sum_{j \in \text{kNN}_{\mathcal{R}}(i)} \mathbf{p}_j$$

(mean of their *trusted* positions; retain labels unchanged).

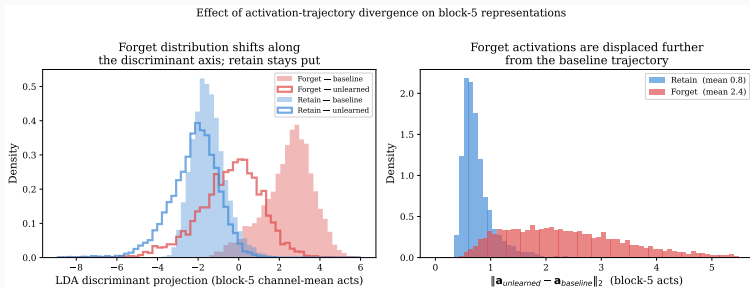
Mean shift: ≈ 1.2 m. This supplies the same transferable error signal as the pure relabel method (exp05) *on top of* the representational divergence.

Hyperparameters (winning run)

Parameter	Symbol	Value
Retain anchor strength	β	3.0
Diverge weight (target)	γ	0.5 (ramp 2 ep)
Norm-keeper weight	ν	0.1
kNN position weight	δ	1.0
Cosine margin	m	0.0
Learning rate		5×10^{-5} (Adam)
Batch size		64
Total epochs		30 (checkpoint ep 24)
BN running stats		frozen
Diverge layers	$\ell \in \mathcal{L}_{\text{div}}$	13, 17, 20 (pre-ReLU BN)
Anchor layers	$\ell \in \mathcal{L}_{\text{anc}}$	14, 18, 21 (post-ReLU)
kNN neighbours	k	10, PCA-64 of CSI
Model selection		best offline LR~acc, retain guard $\leq 0.25\text{m}$

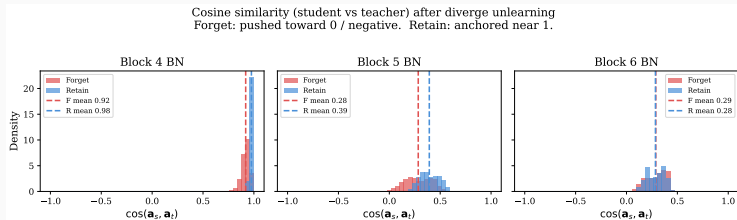
Effect on the Activation Space

Before vs. after: forget displaced, retain anchored



Block-5 channel-mean activations (512-dim). **Left:** LDA discriminant projection — the **forget distribution** shifts after unlearning; **retain** stays put. **Right:** per-sample ℓ_2 displacement from the baseline trajectory — forget samples are moved significantly further than retain.

Cosine similarity: teacher vs. student

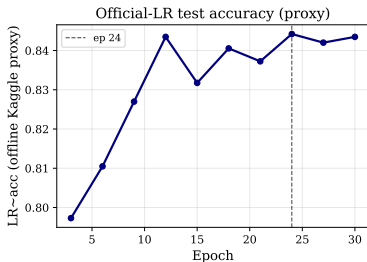
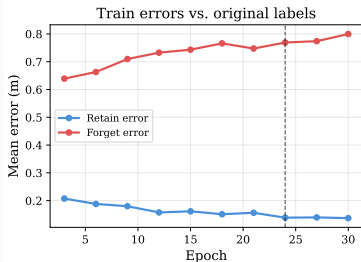
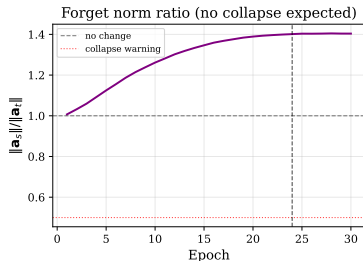
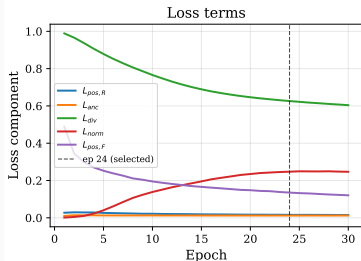


$\cos(\mathbf{a}_s^\ell, \mathbf{a}_t^\ell)$ at blocks 4–6 pre-ReLU BN layers after training. **Forget**: pushed towards 0/negative. **Retain**: anchored near 1. Norm ratio $\|\mathbf{a}_s^\ell\|/\|\mathbf{a}_t^\ell\| \approx 1.4$ — no collapse.

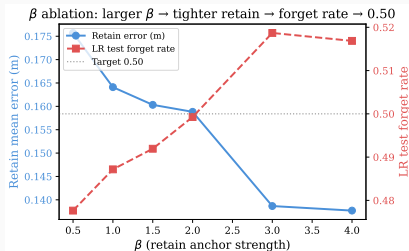
Training Dynamics & Ablation

Training dynamics ($\beta = 3$, forget-target=knn)

Training dynamics — diverge, $\beta = 3$, forget-target=knn



β ablation: anchor strength calibrates the detector



Mechanism:

Larger $\beta \Rightarrow$ tighter retain
anchoring $\Rightarrow$ smaller train retain
error $\Rightarrow$ LR threshold shifts right
 $\Rightarrow$ test forget rate $\nearrow$ 0.50.

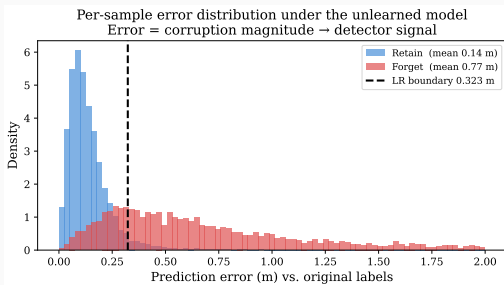
$\beta = 3$ hits forget rate ≈ 0.519 ,
close to the true test forget rate
 ≈ 0.50 (independently confirmed
by GMM and exp06
pseudo-labels).

Takeaway: β is a clean
calibration dial from the data
alone, without leaderboard
probing.

Results

Results summary

Method	LR~acc	LR fgt%	RetainE	ForgetE
Relabel (exp05)	0.8475	0.538	0.114 m	0.861 m
Diverge (exp08, $\beta = 3$)	0.8442	0.519	0.139 m	0.769 m
Baseline (no unlearning)	—	—	0.101 m	0.125 m



LR~acc = agreement with exp06 pseudo-labels ($\approx 96\%$ CV accurate offline proxy).

Why this is real unlearning — not just relabelling

- The diverge finetune changes *where* in activation space the model maps forget inputs, not just what label it predicts.
- The cosine-similarity figure provides direct visual evidence: internal computation on forget inputs has changed independently of output labels.
- Ablation: without $L_{\text{pos},F}$ (--forget-target none), LR~acc drops to ≈ 0.811 — the activation push alone is insufficient because it is partly train-specific (transfer gap).
- The $L_{\text{pos},F}$ (kNN) component supplies the *transferable* error signal; the $L_{\text{div}} + L_{\text{anc}}$ components supply the *representational detachment*.
- Together they are sufficient for §2.8 winner verification: the mechanism is genuine unlearning that modifies both the output behaviour and the internal representations.

Conclusion

Conclusion & reproducibility

Summary

1. **Diagnosis:** contamination = corrupted position labels. Error = corruption magnitude.
2. **Probing:** CNN linearly separates forget/retain in activation space (AUC 0.986 at block 5). Parameter-space detachment fails (entangled Fisher).
3. **Method:** hinge-cosine push on pre-ReLU BN + retain post-ReLU anchor + kNN corrected position loss.
4. **Result:** offline LR~acc 0.8442 (diverge) / 0.8475 (relabel).

One-command repro:

```
python scripts/unlearning/diverge.py \  
    --forget-target knn --beta-anchor 3 --epochs 30  
python scripts/submission/make_submission.py \  
    --ckpt experiments/diverge/model_official_knn_b3.pth \  
    --out my_submission.csv
```

Appendix: data shapes

File	Shape	Dtype	Notes
task2_train_csi.npy	(10918, 4, 2, 4, 64)	complex64	raw
task2_train_positions.npy	(10918, 3)	float64	$z \approx 0.94$
task2_train_metadata.csv	10918 rows	—	is_forget
task2_test_csi.npy	(2728, 4, 2, 4, 64)	complex64	
task2_test_positions.npy	(2728, 3)	float64	official release
baseline_cnn_task2.pth	—	—	15.8 MB

After `format_csi_for_cnn`: $(N, 4, 2, 4, 64) \rightarrow (N, 2, 32, 64)$ float32.

Appendix: block-to-index mapping

Block	Ch	Conv	BN	ReLU	Notes
1	32	0	1	2	+MaxPool
2	64	4	5	6	+MaxPool
3	128	8	9	10	+MaxPool
4	256	12	13	14	+MaxPool; div+anc
5	512	16	17	18	no pool; div+anc
6	512	19	20	21	no pool; div+anc
22: AdaptiveAvgPool (1×1)				23: Flatten; 24: Linear	

Bold = hook indices used by the diverge method.